alive, and we'll find out which when we open the box.

But what does the mathematics say? What does Schrödinger's equation say about Schrödinger's cat? Well, the wave function of the lump of metal is half "radiation was emitted" and half "no radiation was emitted." That interacts with the wave function of the detector, which means they get entangled. So now instead of two wave functions, one for the lump of metal and one for the detector, you have one for the both of them, and now that's in a weird state: half "radiation was emitted and the detector detected it" and half "no radiation was emitted and the detector didn't see anything." As the quantum Rube Goldberg machine continues on its merry way, wave functions keep getting entangled: the wave function of the hammer entangles with the wave function of the detector and metal; the wave function of the vial entangles with the wave function of the hammer and detector and metal, and so on, eventually including the cat itself. The whole system—cat, box, metal, poison, and all—ultimately ends up sharing a single wave function, and that wave function, again, has two equal parts: one part in which radiation was emitted and the cat is dead, and another in which no radiation was emitted and the cat is alive.

So far, so good. Now what happens when you open the box? The usual answer—the answer of the Copenhagen interpretation and the answer given by von Neumann's famous textbook—is that a measurement causes the wave function to collapse. But what if it doesn't? What if we treat you the same way we treated everything in the box? Well, in that case, when you look in the box, you're interacting with it—which means you get entangled with the shared wave function of the box and everything in it. So now we have an even bigger wave function, still with two parts: one where you see a dead cat and a smashed vial of poison, and one where you see a happy cat and an intact vial. Which part of the wave function is real? Everett, remembering Wheeler's advice to take the consequences of physical laws seriously, answered that both are real. There's no way to pick one as more real than the other; the Schrödinger equation treats them equally. So when you perform this experiment, Everett said, both outcomes occur—and you split in two.